

AI Recruit

Innovations, Novelty & Jury Presentation Guide

Automated Resume Screening & Applicant Tracking System
Author: Ramiyaa | Full-stack ATS with explainable NLP (no paid AI API required)

One-Line Summary

AI Recruit is an end-to-end Applicant Tracking System that goes beyond simple resume keyword matching. It combines explainable multi-factor NLP scoring, ethical bias-blind screening, resume integrity verification, JD quality analysis, interview question generation, and a full recruiter workflow in one cloud-deployable platform without requiring any paid AI API.

Previous Problems vs Our Solution (with Uniqueness)

Previous Problem	Our Solution	Uniqueness
Manual resume review takes days for hundreds of applications	Batch AI screening ranks candidates in seconds	Full pipeline, not a single-page upload tool
Inconsistent scoring between recruiters	Weighted formula: 60% Skills + 25% TF-IDF + 15% Profile	Same input gives same output; auditable
Black-box scores with no explanation	Score breakdown + matched/missing skills + AI recommendation	Explainable AI recruiters can trust
Unconscious bias from name, college, gender cues	Bias-Blind Screening Mode anonymizes identity before scoring	DEI-focused ethical hiring feature
Resume fraud and keyword stuffing	Resume Integrity Checker flags unverified skills	Verifies claims, not just keyword overlap
Poor job descriptions produce poor matches	JD Quality Analyzer scores and improves JD before screening	Fixes hiring accuracy at the source
Recruiters unsure what to ask in interviews	Interview Question Generator from skill gaps and profile	Screening extends to interview preparation
Duplicate applications waste recruiter time	Duplicate Detector (email, phone, fuzzy name 88%+)	Real daily recruiter problem most demos ignore
Unstructured PDF/DOCX resumes	Structured parsing: education, projects, internships, certs	Enables profile scoring and integrity checks
Dependence on ChatGPT or paid AI APIs	Custom NLP: TF-IDF, skill dictionary, rule-based engine	Zero API cost; works offline
Scattered tools (Excel, email, folders)	Full ATS: Login to Jobs to Screen to Interview to Analytics	Complete recruiter portal in one product
Dummy login with no real authentication	Supabase Auth + recruiter profiles + session restore	Production-ready secure login
Login page stuck or not visible	Immediate login UI + session timeout + route guards	Reliable auth UX for demo and production
Separate demo and production builds	Demo mode + Live Supabase workspace in one codebase	One app for jury demo and real deployment

Exact Technical Innovations (15 Points)

- Multi-Factor Explainable Scoring: Final = 60% Skill + 25% Similarity + 15% Profile
- Bias-Blind Screening Mode: strips identity; compares standard vs blind score
- Resume Integrity Checker: integrity score 0-100% and risk level low/medium/high
- JD Quality Analyzer: clarity rating and improvement suggestions before screening
- Smart Interview Question Generator: 5-8 questions from gaps and profile
- Duplicate Candidate Detector: email, phone, and fuzzy name matching
- Structured Resume Intelligence: section-aware parsing from PDF/DOCX/TXT
- AI Recommendation Engine: summary, strengths, concerns, action, priority (rule-based)
- End-to-End ATS Workflow: dashboard, jobs, candidates, shortlist, interviews, analytics
- Supabase Real Authentication: email/password, JWT, recruiter_profiles with RLS
- Demo + Production Dual Mode: instant demo or live database workspace
- Batch Screening at Scale: bulk upload, progress tracking, auto re-rank, CSV export
- Side-by-Side Candidate Comparison: 8 metrics with best-value highlighting
- Zero External AI API Dependency: scikit-learn TF-IDF + custom Python logic
- Cloud-Ready Deployment: React + FastAPI + Supabase on Vercel + Render

Scoring Model

Final Score = (60% x Skill Match) + (25% x Resume-JD Similarity) + (15% x Profile Strength)

Skill Match: 50+ skill dictionary + regex overlap with job requirements

Similarity: TF-IDF vectorization + cosine similarity between resume and JD

Profile: education, projects, internships, certifications from structured parsing

Decisions: >= 75% Shortlisted | 45-74% Review | < 45% Not Suitable

Novelty Statement (For Jury / Report)

Traditional resume screening tools either rely on expensive LLM APIs or simple keyword matching, producing opaque scores with no ethical safeguards. AI Recruit introduces a transparent, multi-signal NLP pipeline combined with bias-blind screening, resume integrity verification, JD quality analysis, and interview question generation - all within a production-grade ATS workflow backed by Supabase authentication and PostgreSQL persistence, deployable without any external AI API costs.

30-Second Elevator Pitch

AI Recruit is an end-to-end Applicant Tracking System that automates first-round resume screening using NLP. Recruiters create a job, upload resumes (PDF/DOCX/TXT), and the system extracts skills, compares each resume against the job description using TF-IDF cosine similarity, scores candidates with a transparent weighted model, ranks them, and generates explainable AI recommendations - all inside a full recruiter dashboard with analytics, shortlisting, and interview scheduling.

Quick Jury Q&A

Q: What is the innovation?

A: Explainable, ethical, end-to-end ATS - not just a resume scorer.

Q: Is this just ChatGPT?

A: No. Core engine is custom NLP (TF-IDF, skill dictionary, rules) - no API key.

Q: How is it deployed?

A: Frontend on Vercel, backend on Render, database and auth on Supabase.

Q: Does login work?

A: Yes. Supabase Auth with recruiter profiles; local fallback for demo.

Q: What file formats?

A: PDF, DOCX, and TXT resumes; JD via text or file upload.

Q: How do you ensure fairness?

A: Bias-blind mode + transparent score breakdown + manual override.

Tech Stack

Frontend: React 19, Vite, Tailwind CSS, TypeScript

Backend: Python, FastAPI, Uvicorn, scikit-learn

Database & Auth: Supabase (PostgreSQL + Auth)

Deploy: Vercel (frontend) + Render (backend)

Typical Student Project vs AI Recruit

- One score -> Multi-factor score + breakdown + recommendation
- Keyword only -> Skills + TF-IDF + profile strength
- No explanation -> Matched/missing skills + AI insight card
- No ethics -> Bias-blind screening mode
- No fraud check -> Resume integrity checker
- No interview help -> Auto-generated interview questions
- Single page -> Full recruiter dashboard + ATS workflow
- Needs OpenAI -> Custom NLP, no API key
- No database -> Supabase persistence + real auth
- Not deployed -> Vercel + Render production deployment